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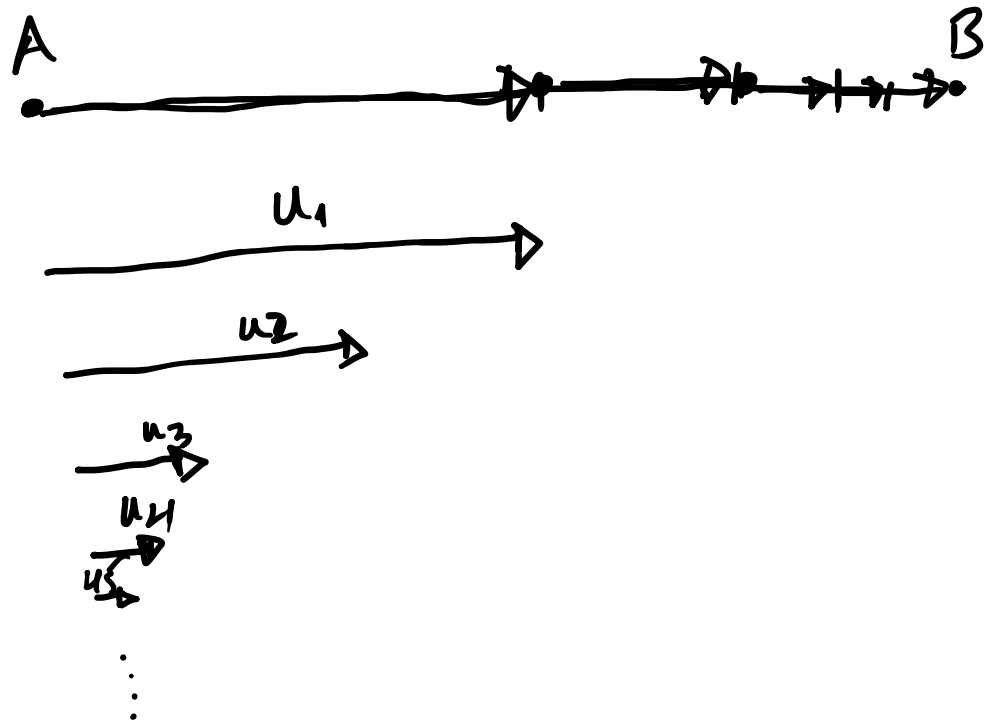


$$u_1 + u_2 + u_3 + \cdots = \overrightarrow{AB}$$

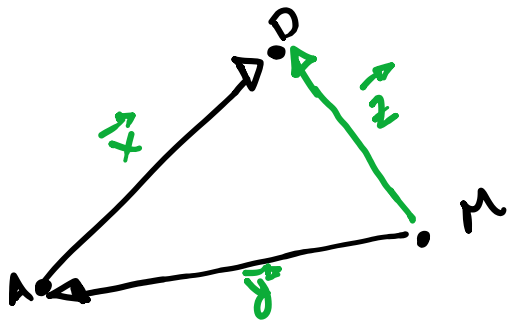
$$S = \underbrace{6+3}_{9} + \underbrace{\left(\frac{3}{2} + \frac{3}{4} + \frac{3}{8} + \frac{3}{16} + \frac{3}{32} + \cdots\right)}_{S_1 \quad PG,\ q = 1/2}$$

Soma da PG inf. é

$$S_1 = \frac{a_1}{1-q}, \quad a_1 = 3/2$$

$$S_1 = \frac{3/2}{1-1/2} = \frac{3/2}{1/2} = \frac{3}{\cancel{2}} \cdot \frac{\cancel{2}}{1} = 3$$

$5 = 9 + 3 = 12$

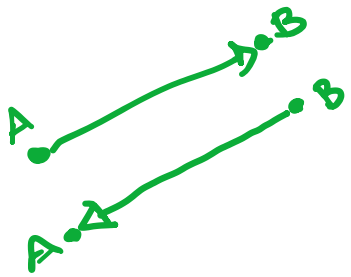


$\vec{z} = \vec{x} - \vec{y}$?

$\vec{z} = \vec{y} - \vec{x}$?

$-\vec{z} = \vec{x} + \vec{y}$

$\vec{z} = -\vec{x} - \vec{y}$



$|\vec{AB}| = |\vec{BA}|$

$\vec{u} = (\overset{x}{3}, \overset{y}{4})$, $\vec{v} = (\overset{x}{7}, \overset{y}{1})$

$\vec{u} + \vec{v} = (3+7, 4+1) = (10, 5)$

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$$\vec{u} = (3,4)$$

$$\alpha = 2.$$

$$\alpha \cdot \vec{u} = 2 \cdot (3,4) = (6,8)$$

.

Exemplo

Determine o vetor $\vec{x}$ que fatisfaz a igualdade

$$3\vec{x} + 2\vec{u} = \frac{1}{2}\vec{v} + \vec{x}$$

onde $\vec{u} = (3,-1)$ e $\vec{v} = (-2,4)$.

$$3\vec{x} + 2 \cdot (3,-1) = \frac{1}{2} \cdot (-2,4) + \vec{x}$$

$$3\vec{x} - \vec{x} + (6,-2) = (-1,2)$$

$$2\vec{x} = (-1,2) - (6,-2)$$

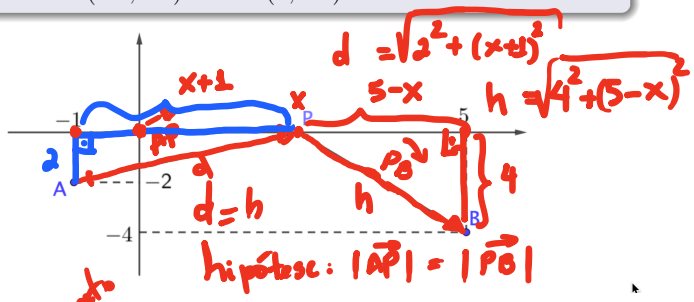
$$2\vec{x} = (-7,4)$$

$$\vec{x} = \frac{1}{2} \cdot (-7,4)$$

$$\boxed{\vec{x} = \left(-\frac{7}{2}, 2\right)}$$

Exercício

Determine o ponto $P = (x, 0)$ no eixo x tal que P é equidistante dos pontos $A = (-1, -2)$ e $B = (5, -4)$.



ponto

$$\vec{AP} = P - A = (x,0) - (-1,-2) = (x+1, 2)$$

$$\vec{PB} = B - P = (5,-4) - (x,0) = (5-x, -4)$$

$$|\vec{AP}| = \sqrt{(x+1)^2 + 2^2} = \sqrt{x^2+2x+5}$$

$$|\vec{PB}| = \sqrt{(5-x)^2 + (-4)^2} = \sqrt{25 - 10x + x^2 + 16}$$

$$= \sqrt{x^2 - 10x + 41}$$

Por hipótese temos

$$\cancel{\sqrt{}}\sqrt{x^2+2x+5} = \cancel{\sqrt{}}\sqrt{x^2-10x+41}$$

$$\cancel{x^2} + 2x + 5 = \cancel{x^2} - 10x + 41$$

$$2x + 10x = 41 - 5$$

$$12x = 36$$

$$x = \frac{36}{12} \Rightarrow \boxed{x = 3}$$